Improving Cross-Lingual Question Answering in African Languages Through Entity-Aware Adaptation

Abstract

Multilingual question answering (QA) systems for low-resourced African languages face a fundamental challenge: while users ask questions in their native languages, the knowledge sources that contain answers exist primarily in high-resource languages like English or French. Current systems typically employ translation-based pipelines—translating queries before retrieval and answers back after extraction—which introduces semantic drift, increases computational cost, and degrades performance on entity-heavy questions. This work investigates whether cross-lingual QA systems can bypass query translation entirely while maintaining or improving retrieval accuracy.

Using the AfriQA benchmark—12,000+ questions across 10 African languages—we systematically compare translation-based pipelines against direct cross-lingual retrieval where multilingual models understand African language queries directly. To address the known weakness of cross-lingual retrieval for entity-centric questions, we propose entity-aware adaptation through multi-task learning with MasakhaNER 2.0, which provides named entity annotations for all 10 target languages. Building on recent work demonstrating that semantic similarity metrics better capture answer correctness in morphologically rich languages, we evaluate models using both traditional metrics (Exact Match, F1) and semantic answer similarity (SAS+).

Preliminary findings from AfriQA establish that cross-lingual retrieval significantly underperforms translation-based methods (19.0% vs. 73.9% recall@10). Our work investigates whether entity-aware training can close this gap, with implications for building more equitable QA technology that serves users in their native languages without compromising answer accuracy.

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1. Background and Motivation

1.1 The African Language Gap in NLP

African languages are spoken by over a billion people yet remain severely underrepresented in natural language processing research and development (Adelani, 2024). While high-resource languages like English enjoy abundant datasets, pre-trained models, and deployed applications, African languages face an "information gap"—native speakers cannot access the vast digital information available online because it exists primarily in colonial languages (Ogundepo et al., 2023). This linguistic exclusion has real-world consequences for education, healthcare, and economic participation.

Dr. David Adelani, a leading researcher in African NLP, notes that recent large language models "produce lower performance for African languages, and there is a large gap in performance compared to high-resource languages for most tasks" (Adelani, 2024). His analysis of models including GPT-4, mT0, and LLaMa 2 across 60 African languages

confirms that performance on African languages "continues to remain a hurdle for the current LLMs, underscoring the need for additional efforts to close this gap."

1.2 Question Answering and the Cross-Lingual Challenge

Question answering systems offer a promising solution: users could ask questions in their native languages while systems retrieve answers from the vast English/French knowledge sources. This is the cross-lingual open-retrieval question answering (XOR QA) paradigm (Asai et al., 2021). However, XOR QA introduces significant technical challenges:

1. Query understanding: Models must comprehend questions in low-resource languages with limited training data
2. Cross-lingual retrieval: Systems must find relevant passages in a different language than the query
3. Answer extraction and translation: Answers must be extracted and returned in the user's language

The AfriQA dataset, created by the Masakhane community in collaboration with Google, represents the first systematic effort to address these challenges for African languages (Ogundepo et al., 2023). Covering 10 geographically and linguistically diverse languages (Bemba, Fon, Hausa, Igbo, Kinyarwanda, Swahili, Twi, Wolof, Yoruba, Zulu), AfriQA provides 12,000+ XOR QA examples where questions are in African languages but answers must be retrieved from English or French Wikipedia.

1.3 The Problem with Translation Pipelines

Current approaches to XOR QA typically rely on translation-based pipelines:

1. Translate the African language query → English/French
2. Retrieve passages using the translated query

3. Extract answer spans from retrieved passages
4. Translate the answer back to the African language

While practical, this approach suffers from several limitations:

Semantic drift: Translation errors compound across the pipeline. Table 4 in the AfriQA paper shows that even state-of-the-art systems like NLLB achieve BLEU scores ranging from 16.6 (Fon) to 53.2 (Swahili)—far from perfect (Ogundepo et al., 2023). Errors in query translation lead to irrelevant retrieval; errors in answer translation produce incorrect responses.

Entity handling: Named entities (people, locations, organizations) are particularly vulnerable in translation pipelines. Proper names may be transliterated inconsistently, and culturally-specific entities may be mistranslated entirely.

Computational cost: Running multiple translation models increases latency and resource requirements—a significant concern for deployment in regions with limited infrastructure.

Code-mixed scenarios: Many African language speakers naturally code-mix with English/French. Translation pipelines struggle with these hybrid inputs.

1.4 The Current State of Research

The AfriQA paper establishes crucial baselines (Ogundepo et al., 2023):

Retrieval Method	Recall@10 (Human Translation)	Recall@10 (No Translation)
BM25	59.4%	—
mDPR	67.6%	—

Hybrid	73.9%	19.0%

Critical finding: Cross-lingual retrieval without translation achieves only 19.0% recall@10—dramatically lower than the 73.9% achieved with human-translated queries. This 55-point gap represents the core challenge your project addresses.

Recent work by Ijebu et al. (2025) identified another crucial issue: evaluation metrics themselves may be underestimating model performance. Their analysis shows that "the lexical dependence of the EM and F1 metrics makes them misjudge predicted answers without lexical overlap"—a particular problem for morphologically rich African languages where the same meaning can be expressed through different word forms. Their proposed SAS+ pipeline (semantic answer similarity enhanced with automatic answer annotation) demonstrated that mT5 and AfroXLMR models actually perform better than previously reported when evaluated semantically.

2. Problem Statement

Despite the creation of AfriQA and initial baselines, three critical gaps remain:

Gap 1: No Systematic Investigation of Entity-Aware Adaptation

The AfriQA paper establishes that cross-lingual retrieval underperforms translation-based methods, but offers no solution for improving it. Meanwhile, MasakhaNER 2.0 provides named entity annotations for all 10 AfriQA languages (Adelani et al., 2022), yet no work has explored whether this NER supervision can improve cross-lingual QA performance. The MasakhaNER 2.0 study demonstrates that "choosing the best transfer language improves zero-shot F1 scores by an average of 14 points across 20 languages compared to using English"—suggesting that careful entity-aware training could yield similar gains for QA.

Gap 2: Evaluation Metrics Mismatch

Ijebu et al. (2025) demonstrate that string-based metrics systematically underestimate performance on morphologically rich languages. However, their work focuses on evaluation only—improving how we measure performance, not improving the models themselves. This leaves open the question: if we both improve models (through entity-aware training) AND evaluate them more accurately (through semantic metrics), how much closer can we get to closing the 55-point retrieval gap?

Gap 3: No Analysis of Entity-Type Sensitivity

It remains unknown whether entity-aware adaptation benefits all entity types equally. Questions involving person names (e.g., "Who was the first president of Zambia?") may benefit more from NER supervision than location-based questions (e.g., "What is the capital of Nigeria?"). Understanding these differences could inform targeted model improvements.

Gap 4: Limited Exploration of Language-Family Transfer

The MasakhaNER 2.0 findings on source language selection—that matching language families improves transfer—have not been applied to QA. Could training on NER from related languages (e.g., Bantu languages together) improve cross-lingual QA more than training on English alone?

3. Research Objectives

This project aims to address these gaps through the following objectives:

Objective 1: Establish Entity-Aware Baselines for AfriQA

We will fine-tune multilingual sequence-to-sequence models (mT5, AfroXLMR) on AfriQA under three conditions:

- Condition A: Standard fine-tuning (baseline)
- Condition B: Multi-task learning: QA + NER simultaneously
- Condition C: Intermediate training: NER first, then QA

We hypothesize that NER-aware training will improve performance on entity-heavy questions, partially closing the gap between cross-lingual and translation-based retrieval.

Objective 2: Implement Semantic Evaluation Framework

Following Ijebu et al. (2025), we will implement the SAS+ pipeline using LaBSE (Language-agnostic BERT Sentence Embedding) to evaluate all model outputs. This will:

- Provide more accurate performance estimates for morphologically rich languages
- Enable comparison between string-based and semantic metrics
- Reveal whether entity-aware gains are masked by traditional evaluation

Objective 3: Analyze Entity-Type and Language-Family Effects

We will tag all AfriQA questions with entities using MasakhaNER 2.0 and analyze:

- Performance by entity type (PER, LOC, ORG)
- Performance by entity density (questions with 0, 1, 2+ entities)
- Transfer effects from NER training on related vs. distant languages

Objective 4: Release Reproducible Artifacts

All code, trained models, and evaluation scripts will be publicly released to enable future research and real-world deployment.

Research Questions

ID	Research Question	Hypothesis
RQ1	Does multi-task learning with MasakhaNER 2.0 improve cross-lingual QA performance on AfriQA compared to standard fine-tuning?	Multi-task learning improves retrieval by 5-10 percentage points
RQ2	Which entity types see the greatest improvement from NER-aware training?	Person entities benefit most; organization entities least
RQ3	Does improvement correlate with entity density in questions?	Strong positive correlation
RQ4	When evaluated with SAS+, do the gains appear larger than with EM/F1 alone?	SAS+ reveals 10-15% higher performance
RQ5	Does NER transfer from related languages outperform English→African transfer?	Language-family transfer yields 5-8% better performance

4. Literature Review

4.1 Core Papers

AfriQA: Cross-lingual Open-Retrieval Question Answering for African Languages

Authors: Ogundepo, O., et al. (2023)
Venue: arXiv preprint
DOI/Link: Available on arXiv

Summary: This paper introduces the first cross-lingual open-retrieval question answering dataset for 10 African languages. The authors collected 12,000+ question-answer pairs where questions are in African languages but answers must be retrieved from English or French Wikipedia. They establish comprehensive baselines using various translation systems (Google Translate, NLLB, M2M-100), retrieval methods (BM25, mDPR, hybrid), and reading comprehension models (mT5, AfroXLMR).

Key findings:

- Human translation achieves 73.9% recall@10; cross-lingual retrieval (no translation) achieves only 19.0%
- Hybrid retrieval (BM25 + mDPR) outperforms either method alone
- NLLB provides broad language coverage but lower quality than Google Translate
- The dataset is released under CC BY 4.0 and available on Hugging Face

Relevance to your work: Primary dataset source. Provides the problem definition and baseline numbers you aim to improve.

Cross-Encoder-Based Semantic Evaluation of Extractive and Generative Question Answering in Low-Resourced African Languages

Authors: Ijebu, F. F., Liu, Y., Sun, C., Jere, N. R., Mienye, I. D., & Inyang, U. G. (2025)
Venue: Technologies, 13(3), 119
DOI: 10.3390/technologies13030119

Summary: This paper identifies a fundamental flaw in evaluating African language QA systems: string-based metrics (EM/F1) systematically underestimate performance on morphologically rich languages where the same meaning can be expressed through different word forms. The authors propose SAS+ (Semantic Answer Similarity), a pipeline using LaBSE embeddings combined with automatic answer annotation to evaluate QA outputs semantically rather than lexically.

Key findings:

- Traditional EM/F1 misjudge answers without lexical overlap
- SAS+ reveals mT5 and AfroXLMR perform better than previously reported
- The pipeline is "more efficient than the prevailing string-based F1 and Exact Match metrics" and provides "a more natural performance estimator"
- Models show stronger performance on generative QA than extractive QA when evaluated semantically

Relevance to your work: Provides the evaluation framework. Your work will use SAS+ to more accurately measure entity-aware gains.

MasakhaNER 2.0: Africa-centric Transfer Learning for Named Entity Recognition

Authors: Adelani, D. I., et al. (2022)
Venue: Proceedings of EMNLP 2022
DOI/Link: TIB DOI: 10.57702/wb447t4o

Summary: This paper presents the largest manually annotated NER dataset for 20 African languages, covering all 10 AfriQA languages plus 10 more. The authors systematically study transfer learning for NER, comparing zero-shot transfer from various source languages and fine-tuning strategies.

Key findings:

- Choosing the best transfer language improves zero-shot F1 by 14 points on average compared to using English
- Languages within the same family (e.g., Bantu) transfer better to each other
- Multi-task learning with multiple source languages yields the best performance
- Dataset includes PER, LOC, ORG, and DATE annotations

Relevance to your work: Provides the entity supervision for multi-task learning. The transfer findings inform RQ5 on language-family effects.

How Good are Large Language Models on African Languages?

Authors: Adelani, D. I. (2024)
Venue: EAAMO Colloquium
DOI/Link: https://www.youtube.com/watch?v=RraWCrIK_-U

Summary: This comprehensive evaluation tests models including GPT-4, GPT-3.5, mT0, and LLaMa 2 across 60 African languages and 6 tasks (NER, news classification, sentiment analysis, machine translation, QA, and semantic similarity). The study provides the most extensive benchmark of LLM performance on African languages to date.

Key findings:

- mT0 performs best for cross-lingual QA among all tested models
- There remains "a large gap in performance compared to high-resource languages"
- GPT-4 shows strong zero-shot capabilities but lags on low-resource languages
- Performance varies dramatically across languages and tasks

Relevance to your work: Supports model selection (mT5/mT0 as base models) and confirms the ongoing need for improved African language QA systems.

SERENGETI: Massively Multilingual Language Models for Africa

Authors: UBC-NLP (2023)
Venue: GitHub repository
DOI/Link: https://github.com/UBC-NLP/serengeti

Summary: SERENGETI is a collection of massively multilingual language models pretrained on 517 African languages and 16 high-resource languages. The models are evaluated on the AfroNLU benchmark, which includes MasakhaNER, MasakhaPOS, and AfriSenti among other tasks.

Key findings:

- SERENGETI outperforms mBERT and XLMR on MasakhaNER and MasakhaPOS
- Pretraining on more African languages improves downstream performance
- Models are available in small, base, and large configurations
- The AfroNLU benchmark provides comprehensive evaluation across 32 languages

Relevance to your work: Alternative model choice. SERENGETI may be stronger than mT5/AfroXLMR for entity-aware adaptation given its extensive African language pretraining.

InkubaLM: A small language model for low-resource African languages

Authors: Tonja, A. L., et al. (2024)
Venue: arXiv preprint arXiv:2408.17024
DOI/Link: https://arxiv.org/abs/2408.17024

Summary: This paper presents InkubaLM, a 0.4B parameter language model pretrained on 5 major African languages (Hausa, Yoruba, Swahili, isiZulu, isiXhosa). The model is evaluated on the Inkuba-Instruct dataset, which includes QA, NER, and POS tasks.

Key findings:

- InkubaLM achieves competitive performance with models 10x its size
- The model is designed for efficient deployment in resource-constrained settings
- Inkuba-Instruct provides 100M+ instruction samples across 5 languages
- Shows strong performance on entity recognition despite small size

Relevance to your work: Potential lightweight alternative for deployment-focused experiments. The Inkuba-Instruct dataset could supplement MasakhaNER for languages it covers.

4.2 Summary of Key Gaps

Paper	What They Did	What's Missing
AfriQA (2023)	Created dataset; established baselines	No attempt to improve cross-lingual retrieval; no NER integration
Ijebu et al. (2025)	Proposed semantic evaluation	Focused only on evaluation, not model improvement
MasakhaNER 2.0 (2022)	Created NER dataset; studied NER transfer	Applied to NER only, not QA

Adelani (2024)	Evaluated LLMs on African languages	Evaluation only, no adaptation

Your contribution: First systematic investigation combining entity-aware model improvement WITH semantic evaluation for African language QA.

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5. Datasets

5.1 Primary Dataset: AfriQA

Description: Cross-lingual open-retrieval QA dataset with questions in 10 African languages and answers in English/French (Ogundepo et al., 2023).

Access: Hugging Face - `masakhane/afriqa-gold-passages`

Statistics:

Language	# Questions	Pivot Language
Bemba	[Check dataset]	English
Fon	[Check dataset]	French
Hausa	[Check dataset]	English
Igbo	[Check dataset]	English
Kinyarwanda	[Check dataset]	English

Swahili	[Check dataset]	English
Twi	[Check dataset]	English
Wolof	[Check dataset]	French
Yoruba	[Check dataset]	English
Zulu	[Check dataset]	English
Total	12,239	

Split sizes: Train (4,333), Dev (4,346), Test (3,560)

Format:

```json
{
  "id": "unique_identifier",
  "question": "Question in African language",
  "question_language": "Language code (e.g., 'yor')",
  "answer": "Answer in African language",
  "answer_pivot": "Answer in pivot language (English/French)",
  "passage": "Relevant passage in pivot language",
  "passage_id": "Wikipedia passage identifier"
}
```

5.2 NER Dataset: MasakhaNER 2.0

Description: Named entity annotations for 20 African languages, covering all AfriQA languages (Adelani et al., 2022).

Access: TIB DOI - `10.57702/wb447t4o` or Hugging Face - `masakhane/masakhaner2`

Languages covered in AfriQA:

Language	Entity Types Annotated
Bemba	PER, LOC, ORG, DATE
Fon	PER, LOC, ORG, DATE
Hausa	PER, LOC, ORG, DATE
Igbo	PER, LOC, ORG, DATE
Kinyarwanda	PER, LOC, ORG, DATE
Swahili	PER, LOC, ORG, DATE
Twi	PER, LOC, ORG, DATE
Wolof	PER, LOC, ORG, DATE
Yoruba	PER, LOC, ORG, DATE

Zulu	PER, LOC, ORG, DATE

Format:

```json
{
  "id": "sentence_id",
  "text": "Sentence in African language",
  "language": "Language code",
  "entities": [
    {
      "text": "Entity text",
      "type": "PER/LOC/ORG/DATE",
      "start": 0,
      "end": 5
    }
  ]
}
```

5.3 Evaluation Dataset: SAS+ Implementation Resources

Description: Ijebu et al. (2025) provide code and methodology for semantic evaluation. No separate dataset is needed; you will generate predictions and evaluate using their pipeline.

Access: The paper includes implementation details; code may be available from authors.

Requirements:

- LaBSE model (Hugging Face: `sentence-transformers/LaBSE`)
- Predicted answers from your models
- Ground truth answers from AfriQA

5.4 Supplementary Datasets (Optional)

Inkuba-Instruct

Description: Instruction dataset covering Hausa, Yoruba, Swahili, isiZulu, isiXhosa with QA, NER, and POS tasks (Tonja et al., 2024)
Access: Hugging Face - `lelapa/lelapa_instruct_datasets`
Use case: Could supplement MasakhaNER for the 5 languages it covers

AfroNLU Benchmark

Description: Evaluation suite for African language understanding, included with SERENGETI models (UBC-NLP, 2023)

Access: GitHub - `UBC-NLP/serengeti`

Use case: Additional evaluation if using SERENGETI models

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6. Experimental Design

6.1 Models

Model	Size	Why Use It	Access
mT5-base	580M	Used in AfriQA baselines; strong generative QA	Hugging Face: `google/mt5-base`
AfroXLMR-base	270M	Used in Ijebu et al.; strong on African NLU	Hugging Face: `Davlan/afroxmlr-base`
SERENGETI	Varies	Pretrained on 517 African languages; may outperform others	Hugging Face: `UBC-NLP/serengeti-base`
mT0	Various	Best for cross-lingual QA per Adelani (2024)	Hugging Face: `bigscience/mt0-base`

6.2 Training Conditions

Condition ID	Description	Purpose
A	Standard fine-tuning on AfriQA only	Replicate AfriQA baseline
B	Multi-task: AfriQA + MasakhaNER 2.0 (joint training)	Test if simultaneous learning helps
C	Intermediate: Train on MasakhaNER 2.0 → fine-tune on AfriQA	Test sequential transfer
D	Adapter-based: Inject NER adapters into frozen QA model	Parameter-efficient approach
E	Language-family transfer: Train NER on related languages → fine-tune on AfriQA	Test cross-lingual transfer hypothesis

6.3 Training Configuration

Hyperparameters (based on AfriQA paper):

Parameter	Value
Learning rate	3e-5
Batch size	16
Epochs	5-10 (early stopping)

Warmup steps	500
Weight decay	0.01
Max sequence length	256
Gradient accumulation	2
Multi-task weight (α)	0.1, 0.3, 0.5 (tune)
Random seeds	42, 123, 456

Computational requirements:

- 4x A100 GPUs (or equivalent) for full experiments
- 1x A100 for pilot experiments on 2-3 languages
- Estimated runtime: 2-3 days for full training runs

6.4 Evaluation Metrics

Metric	Description	Implementation
Exact Match (EM)	String match with ground truth	For comparability with AfriQA
Token-level F1	Token overlap	For comparability with AfriQA
SAS+	Semantic similarity using LaBSE embeddings	Follow Ijebu et al. 2025

Entity-F1	F1 specifically on entity spans	Custom implementation

SAS+ Implementation Steps:

1. Generate predicted answer spans from your models
2. Encode both predicted and ground truth answers using LaBSE
3. Compute cosine similarity between embeddings
4. Apply optimal threshold (determined on dev set)
5. Calculate semantic accuracy (percentage above threshold)

6.5 Entity Analysis Framework

Step 1: Tag AfriQA questions with entities

Run MasakhaNER 2.0 on all AfriQA questions to identify:

- Person mentions (PER)
- Location mentions (LOC)
- Organization mentions (ORG)
- Date mentions (DATE)

Step 2: Create entity-based subsets

Subset	Definition

PER-questions	Questions containing at least one person entity
LOC-questions	Questions containing at least one location entity
ORG-questions	Questions containing at least one organization entity
No-entity questions	Questions containing no entities
Low-density	Questions with 0-1 entities
High-density	Questions with 2+ entities

Step 3: Stratified evaluation

Evaluate each condition on each subset separately to identify:

- Which entity types benefit most from NER-aware training
- Whether high-density questions show larger improvements

6.6 Statistical Validation

To ensure results are robust:

- Run each condition with 3 different random seeds
- Report mean and standard deviation
- Use paired bootstrap tests for significance ($p < 0.05$)
- Apply Bonferroni correction for multiple comparisons

6.7 Code Structure (Pseudo)

```python
# multi_task_qa_ner.py

import torch
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer
from datasets import load_dataset

class MultiTaskQANER(torch.nn.Module):
    def __init__(self, model_name):
        super().__init__()
        self.base_model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)

        # NER head (if using multi-task architecture)
        self.ner_head = torch.nn.Linear(
            self.base_model.config.hidden_size,
            4  # PER, LOC, ORG, DATE
        )

    def forward(self, input_ids, attention_mask, task, labels=None):
        encoder_outputs = self.base_model.encoder(input_ids, attention_mask)

        if task == "qa":
            decoder_outputs = self.base_model.decoder(
                input_ids=labels,
                encoder_hidden_states=encoder_outputs
            )
            return self.base_model.lm_head(decoder_outputs)
```

```python
        elif task == "ner":
            sequence_output = encoder_outputs.last_hidden_state
            return self.ner_head(sequence_output)

# Training loop
def train_epoch(model, dataloader, optimizer, alpha=0.3):
    total_loss = 0

    for batch in dataloader:
        optimizer.zero_grad()

        if batch["task"] == "qa":
            outputs = model(
                input_ids=batch["input_ids"],
                attention_mask=batch["attention_mask"],
                task="qa",
                labels=batch["labels"]
            )
            loss_qa = outputs.loss
            loss = loss_qa

        elif batch["task"] == "ner":
            outputs = model(
                input_ids=batch["input_ids"],
                attention_mask=batch["attention_mask"],
                task="ner"
            )
            loss_ner = torch.nn.functional.cross_entropy(
                outputs.view(-1, 4),
                batch["ner_labels"].view(-1)
            )
            loss = alpha * loss_ner
```

```python
        loss.backward()
        optimizer.step()
        total_loss += loss.item()

    return total_loss / len(dataloader)

# Evaluation with SAS+
def evaluate_sas(model, test_data, tokenizer, threshold=0.8):
    from sentence_transformers import SentenceTransformer
    laBSE = SentenceTransformer('sentence-transformers/LaBSE')

    predictions = []
    ground_truths = []

    for example in test_data:
        # Generate prediction
        pred = model.generate(example["question"])
        predictions.append(pred)
        ground_truths.append(example["answer"])

    # Encode
    pred_embeds = laBSE.encode(predictions)
    truth_embeds = laBSE.encode(ground_truths)

    # Compute similarities
    similarities = torch.cosine_similarity(pred_embeds, truth_embeds)

    # Calculate semantic accuracy
    semantic_acc = (similarities > threshold).float().mean()


    return semantic_acc.item()
```

7. Expected Contributions

7.1 Summary of Contributions

Contribution	Novelty
First entity-aware adaptation for African language QA	No prior work combines MasakhaNER + AfriQA
Combined model improvement + semantic evaluation	Ijebu et al. did evaluation only; we improve models AND evaluate better
Entity-type and language-family analysis	First systematic analysis of which entities benefit from NER supervision
Publicly available entity-tagged AfriQA subset	Enables future research on entity-centric QA
Reproducible code and trained models	Community resource for deployment

7.2 Expected Outcomes by Research Question

RQ	Expected Finding	Evidence from Literature

RQ1	Multi-task learning improves retrieval by 5-10%	MasakhaNER 2.0 showed 14-point gains for NER transfer
RQ2	Person entities benefit most	Persons have most varied surface forms; NER helps normalize
RQ3	Strong positive correlation	Entity-dense questions require entity awareness
RQ4	SAS+ reveals 10-15% higher performance	Ijebu et al. found string-based metrics underestimate
RQ5	Language-family transfer yields 5-8% gains	MasakhaNER 2.0 transfer findings

7.3 Hypothetical Results Table (Illustrative)

Condition	EM	F1	SAS+	Entity-F1 (PER)	Entity-F1 (LOC)
A (Baseline)	32.4	42.4	47.8	38.2	41.5
B (Multi-task)	37.8	48.2	58.3	49.7	48.2
C (Intermediate)	38.2	48.9	59.1	51.2	48.9
D (Adapters)	35.1	45.3	52.4	43.8	44.1
E (Family transfer)	39.5	50.2	61.4	53.5	51.7

Note: These are illustrative numbers based on literature, not actual results.

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8. Timeline

Phase	Duration	Activities	Deliverables
Phase 1: Setup	Weeks 1-3	Download datasets; set up environment; reproduce AfriQA baselines	Working environment; reproduced baselines
Phase 2: Pilot	Weeks 4-6	Implement multi-task learning on 3 languages; tune hyperparameters	Working code; optimal hyperparameters
Phase 3: Full experiments	Weeks 7-10	Run all conditions on all 10 languages; collect raw outputs	Raw predictions; trained models
Phase 4: Analysis	Weeks 11-13	Entity tagging; statistical analysis; SAS+ evaluation	Analysis scripts; result tables
Phase 5: Writing	Weeks 14-16	Paper writing; code release; visualizations	Draft paper; GitHub repo

Detailed weekly breakdown:

Week 1-2: Download and explore AfriQA and MasakhaNER 2.0; set up Python environment with Hugging Face

Week 3: Reproduce AfriQA baseline results on 2-3 languages to validate setup

Week 4: Implement multi-task learning architecture; test on 1 language

Week 5: Tune hyperparameters (α weighting, learning rate) on pilot languages

Week 6: Finalize pilot experiments; document best configurations

Week 7-8: Run Conditions A, B, C on all 10 languages

Week 9: Run Conditions D, E on all 10 languages

Week 10: Verify all runs complete; collect and organize outputs

Week 11: Implement entity tagging pipeline; tag all AfriQA questions

Week 12: Run subset evaluations (by entity type, density)

Week 13: Implement SAS+ evaluation; run on all predictions

Week 14: Statistical analysis; create result tables and figures

Week 15: Draft paper; write related work and methodology sections

Week 16: Finalize paper; prepare code release; create visualizations

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9. References

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Appendix: Quick Start Guide

A.1 Environment Setup

```bash
# Create conda environment
conda create -n afriqa-ner python=3.9
conda activate afriqa-ner

# Install dependencies
pip install torch torchvision torchaudio
```

```
pip install transformers datasets sentence-transformers
pip install evaluate scikit-learn pandas numpy
pip install jupyter matplotlib seaborn

# Clone repositories (if needed)
git clone https://github.com/masakhane-io/masakhane-ner
```

```
git clone https://github.com/UBC-NLP/serengeti
```

A.2 Download Datasets

```
python

# Python script to download datasets
from datasets import load_dataset

# AfriQA
afriqa = load_dataset("masakhane/afriqa-gold-passages")
print(afriqa)
```

```
# MasakhaNER
```

Continue